Question	Answer	Marks
4(a)	The e.m.f. of accumulator is the sum of p.d.s across the external resistor and the resistance wire. e.m.f. = 8.00 V.	**A1**
4(b)	Null deflection means the no current flows through the unknown cell. Thus, terminal p.d across unknown cell = p.d. across AB terminal p.d. across unknown cell = 4.00 V e.m.f. of unknown cell = terminal p.d across unknown cell = 4.00 V	**M1** **A1**
4(c)(i)	p.d. across AC = (e.m.f. of accumulator) [(72.0/120.0) R_1 / (72.0/120.0) R_1 + R_1] = 3.00 V	**M1** **A1**
4(c)(ii)	p.d. across R_2 = (p.d. across AC) = 3.00 V	**A1**
4(c)(iii)	4.00 X 12.0 / (r + 12.0) = 3.00 r = 4.0 Ω	**M1** **A1**
	Max Marks	**8**

Setter: Koon Loon Marker:		
Question	**Answer**	**Marks**
5(a)(i)	Square loop rotate about axis CD	**A1**